

KOHLER

AquaSense OS

Real-time hydro-intelligence for commercial facilities
Based on Pune International Airport (PNQ)- NITB

-Anvi Singh Parihar

5.5% – 6.5%

of commercial water lost to undetected micro-leaks

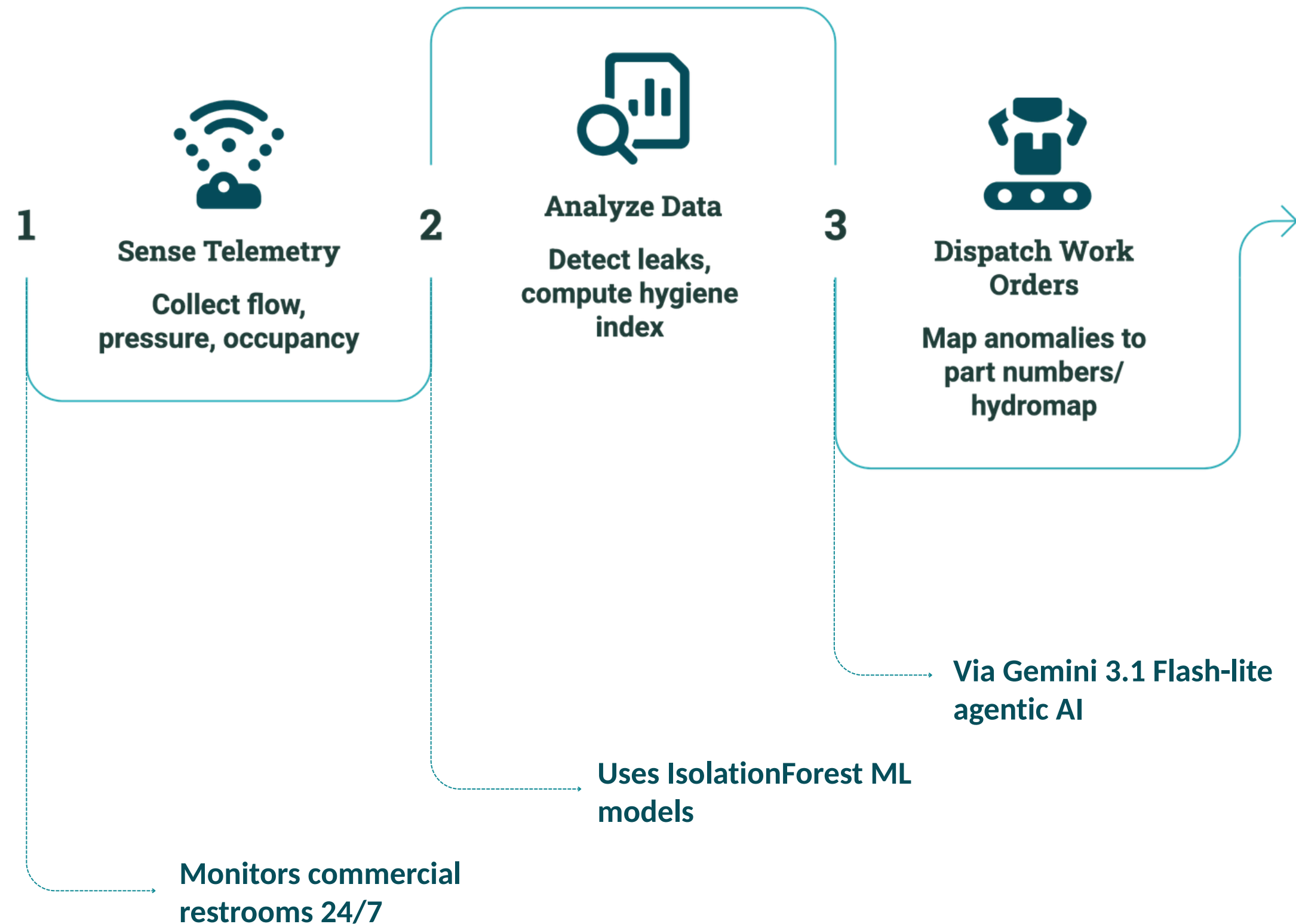
72 hrs

average fault-to-detection delay without intelligence

₹18.5B/yr

commercial non-weather water damage cost

CORE APPROACH



SYSTEM ARCHITECTURE

IOT LAYER

Smart fixtures' data

INGEST LAYER

validates raw
telemetry payloads
instantly and buffers
streams → no
packet loss

ANALYTICS LAYER

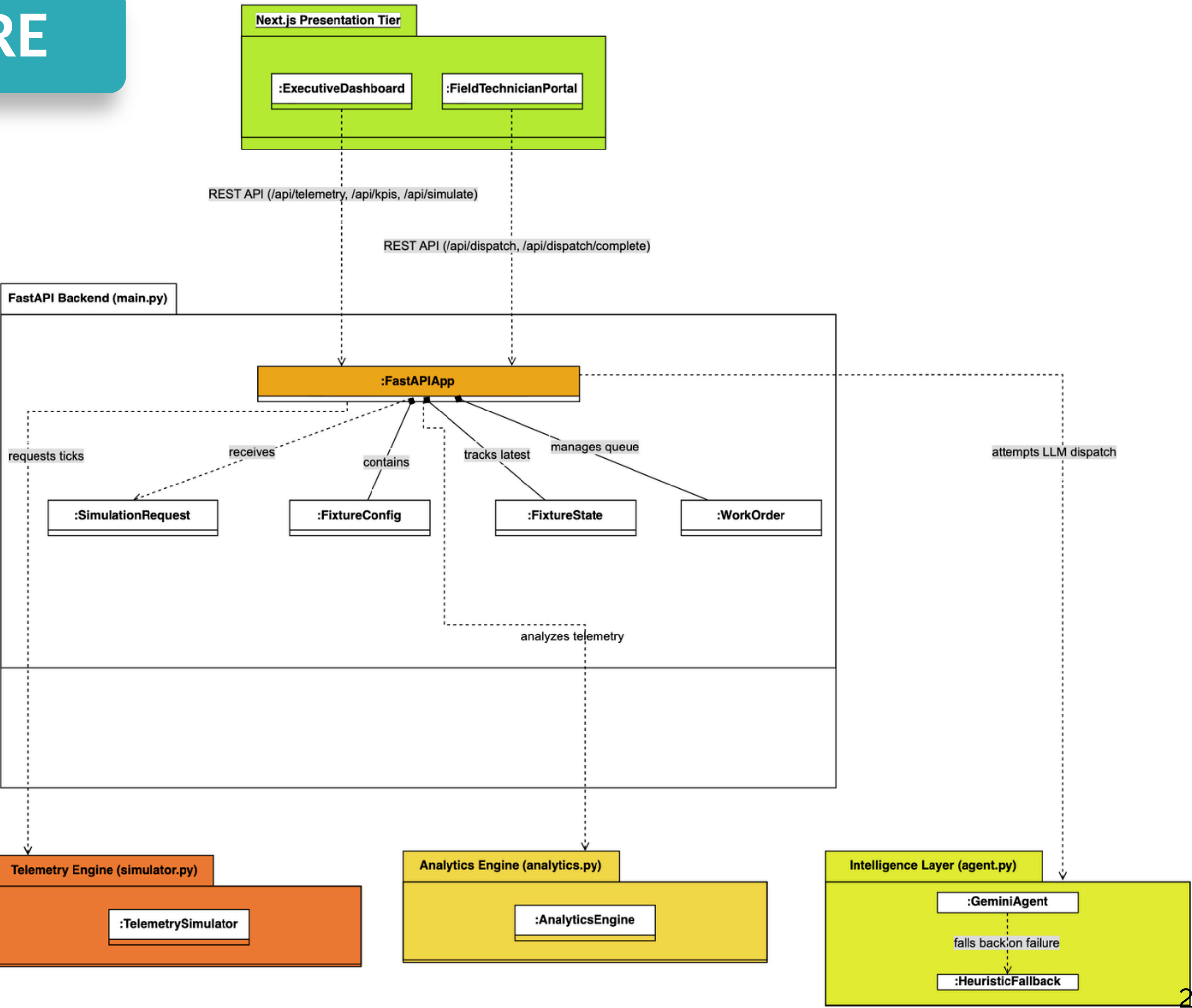
Runs statistical
calculations and an
IsolationForest ML
model to flag anomalies

AGENT LAYER

Passes the anomalies to
Gemini 3.1 Flash-lite,
which determines root
causes and matches the
exact replacement
parts.

FRONTEND LAYER

Renders live and dynamic
data across the Executive
Dashboard and Technician
Portal.



TECH STACK

FRONTEND

Next.js & React
Tailwind CSS

BACKEND

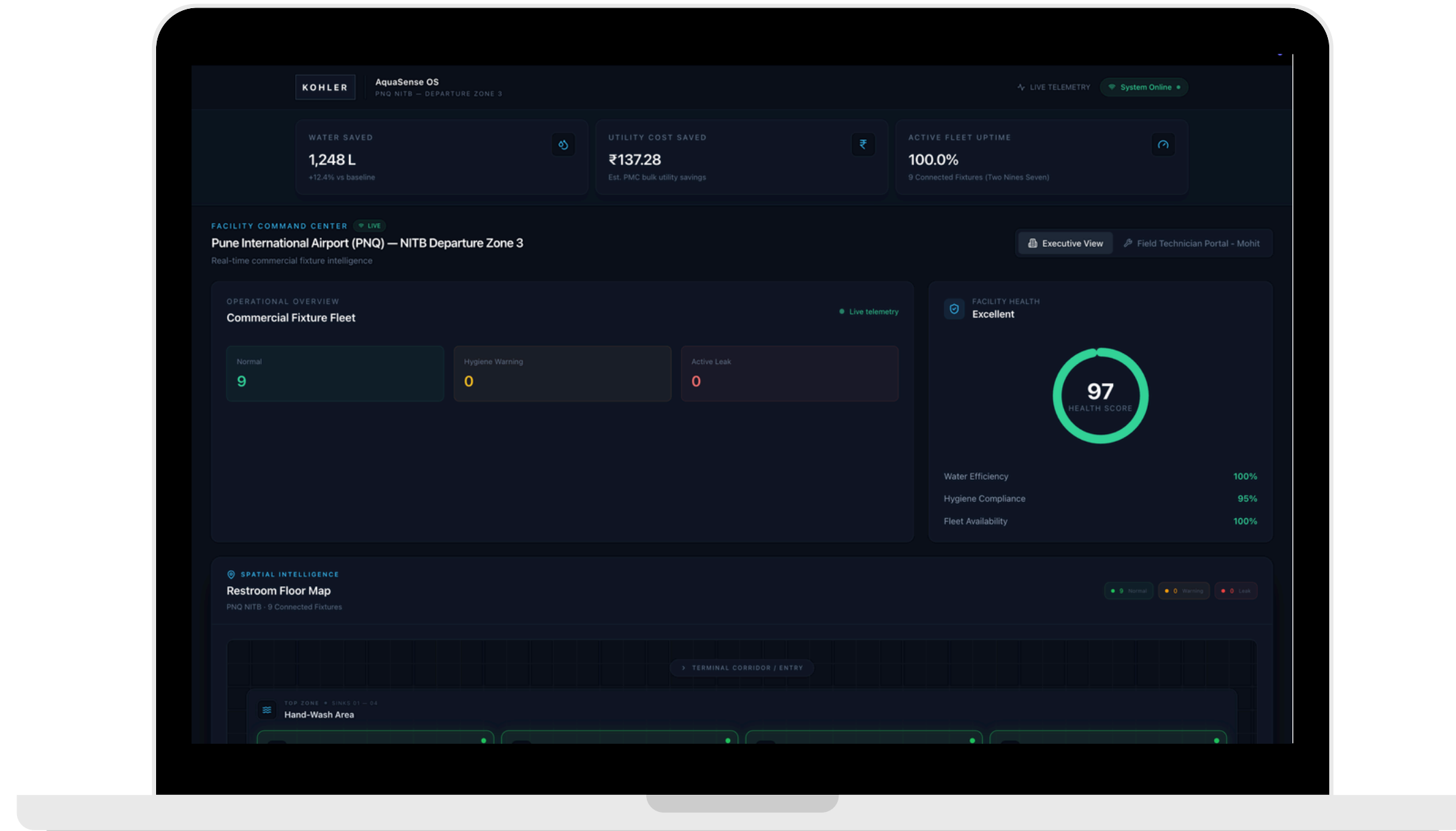
Python & FastAPI
Pydantic

ML & ANALYTICS

Scikit-learn IsolationForest
Rolling Statistical Analysis
Dynamic Hygiene Engine

AGENTIC AI

Gemini 3.1 Flash-lite
Strict JSON Enforcement
Deterministic Fallback Engine



INNOVATION PITCH

				
Metric	Work Order JSON Parse Rate	Exact-SKU Matching	Anomaly → Dispatch End-to-End Latency	Reduction in Mean Time to Repair
Value	100%	40%	< 3 sec	~75% - 80%

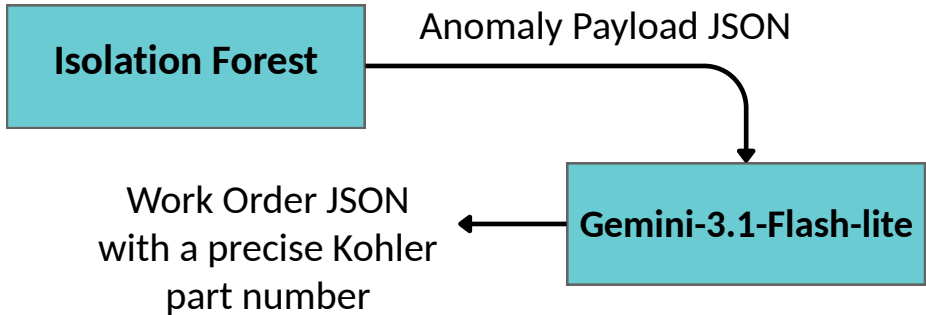
WATER STEWARDSHIP

- Spatial Hydro-Intelligence
- Continuous 24/7 monitoring
- Hours of log analysis → one glance look

60 L/h water saved per resolved CRITICAL fault

KOHLER ECOSYSTEM VALUE

Agentic AI Parts Dispatch



40% Exact-SKU matching

OPERATIONAL EFFICIENCY

- Realtime UI Sync
- Instant KPI Updates
- Zero Manual Efforts- Eliminates stale dashboards

Zero manual state sync between any 2 components